



Smart Medicine Reminder System for DVAGO

Final Year Project Report

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Hamdard Institute of Engineering and Technology
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Certificate of Approval



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This project “**Smart Medicine Reminder System for DVAGO**” is presented by **Abdul Ghani, Syed Anas Ali** under the supervision of their project advisor and approved by the project examination committee, and acknowledged by the Hamdard Institute of Engineering and Technology, in fulfillment of the requirements for the Bachelor’s degree in Computer Science.

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Authors' Declaration

We declare that this project report was carried out in accordance with the rules and regulations of Hamdard University. The work is original except where indicated by special references in the text and no part of the report has been submitted for any other degree. The report has not been presented to any other University for examination.

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We, Abdul Ghani and Syed Anas Ali, solemnly declare that the work presented in this Final Year Project Report, Smart Medicine Reminder System for DVAGO, has been carried out solely by ourselves with no significant help from any other person except where such help has been duly acknowledged. We confirm that no portion of this report has been plagiarized and that any material drawn from other sources has been properly referenced.

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Acknowledgment

All praises and thanks be to Almighty **ALLAH**, the Most Gracious, the Most Merciful, the Lord of all worlds, the Sustainer and the Source of all Knowledge, Wisdom, and Strength. Without His countless blessings and divine guidance, the successful completion of this project would not have been possible. All respects and eternal salutations to the Last Prophet, **Hazrat MUHAMMAD** ﷺ, the Mercy to mankind, whose Exemplary Life, Character, and Teachings continue to enlighten and guide the entire humanity. We also extend our utmost love and reverence to his Noble AAL (family), The Ahl-e-Bayt, whose piety, sacrifices, and unwavering dedication to the message of Islam serve as a timeless inspiration for us in all walks of life. May **Allah**'s peace and blessings be upon them all. We would like to express our sincere gratitude to the **Hamdard University**, (HIET) for providing us with an academic environment full of encouragement, resources, and the right guidance throughout the journey of this project. It is indeed a matter of great honor and pride for us to complete this project under the banner of such a prestigious institution, where excellence, ethics, and education walk hand in hand. Our heartfelt appreciation goes to our respected supervisor, **Mr. Muhammad Salman**, whose unwavering support, expert guidance, and constant encouragement were instrumental in shaping our work. His valuable insights, technical expertise, and motivating feedback helped us overcome challenges and stay focused on our goal. We are also sincerely thankful to **Mr. Maaz Ahmed**, our Co-Supervisor, and to our external supervisor, **Mr. Muhammad Hamza**, whose perspective from outside the academic setting helped us think more practically about real-world usability and deployment concerns for his continuous support, guidance, and valuable suggestions during the course of this project. We are truly grateful to our teachers and academic mentors who laid the foundation of knowledge and whose teachings and values guided us throughout our academic life. Each lecture, every project discussion, and classroom interaction added layers to our understanding and development. We would like to extend special thanks to our classmates, batchmates, and project team members for their cooperative spirit, valuable suggestions, constructive criticism, and for fostering a collaborative environment. Their encouragement during times of difficulty and celebration of milestones helped us remain motivated. We cannot forget to mention the boundless love, support, and prayers of our beloved parents and families. Their sacrifices, moral strength, and continuous encouragement have been our biggest pillars of support. Their belief in our potential gave us the confidence to overcome every obstacle and accomplish our academic and personal goals. Lastly, we would like to thank all those individuals, whether named or unnamed, who helped us directly or indirectly in the successful completion of this project. Every act of support, no matter how small, played a significant role in making this achievement possible. May **Allah** bless each one of them with the best of rewards in both this world and the Hereafter.

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Definition of Terms, Acronyms, and Abbreviations

Table 2: Definition of Terms, Acronyms and Abbreviations

Term	Description
AI	Artificial Intelligence
API	Application Programming Interface
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
CPU	Central Processing Unit
DB	Database
FCM	Firebase Cloud Messaging
FYP	Final Year Project
GUI	Graphical User Interface
GPU	Graphics Processing Unit
HTML	Hyper Text Markup Language
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol Secure
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
JS	JavaScript

Term	Description
MongoDB	Mongo Database
NoSQL	Not Only Structured Query Language
OCR	Optical Character Recognition
OS	Operating System
PDF	Portable Document Format
REST	Representational State Transfer
SDS	Software Design Specifications
SRS	Software Requirements Specifications
SSL	Secure Sockets Layer
TLS	Transport Layer Security
TTS	Text-to-Speech
UI	User Interface
UX	User Experience
UML	Unified Modeling Language

Abstract

Missing or mistimed medicine doses are a common problem, and for most people it has little to do with carelessness and everything to do with a busy day, a forgotten alarm, or simply not having a dependable way to be reminded. The Smart Medicine Reminder System for DVAGO is a mobile application built to close that gap. It gives users a straightforward way to set up medicine reminders, receive timely push notifications and voice alerts, and even pull medicine details directly from a photo of the packaging using AI-assisted optical character recognition, rather than typing everything in by hand.

The application is built around a layered architecture, with a mobile front end, a set of backend services, and Firebase handling authentication and notification delivery in the background. A MongoDB Atlas database stores user profiles, reminder schedules, and scan history, keeping everything synchronized across sessions and devices. The reminder engine itself is designed to be reliable under everyday conditions: notifications fire close to their scheduled time, dosage history is logged automatically, and the interface stays simple enough that someone who isn't particularly comfortable with technology can still use it without much friction.

Development followed an Agile, incremental approach, with each core module, authentication, reminder scheduling, AI scanning, and notification delivery, built and tested as a separate, working piece before being integrated into the larger system. This report walks through the reasoning behind the system's design, the technical decisions that shaped its architecture, and the results of testing it against real usage scenarios, with the aim of showing how a relatively simple combination of scheduling, automation, and AI can make a meaningful difference in how consistently people take their medicine.

Keywords:

- Medication Adherence
- Mobile Health Application
- Medicine Reminder
- Push Notifications
- Voice Alerts
- Optical Character Recognition (OCR)
- Firebase Cloud Messaging
- AI-Assisted Medicine Recognition
- Healthcare Technology

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CHAPTER 1

INTRODUCTION

1.1 Motivation

Keeping up with a medicine schedule sounds simple until it actually has to happen every single day, often more than once, sometimes for weeks or months at a stretch. In practice, people forget. A dose gets pushed back because of a meeting, a child needs attention, or the strip of tablets is simply not within reach at the right moment. None of this happens because someone is careless about their health; it happens because remembering a recurring task with no built-in reminder is genuinely hard to sustain over time.

DVAGO, as a digital pharmacy platform, already gives users a way to order and manage their medicines online, but it does not yet offer anything that helps them remember to actually take what they have ordered. That gap is what motivated this project. The Smart Medicine Reminder System for DVAGO was built to sit alongside that existing platform and address the part of the medication journey that happens after the purchase: making sure the right dose gets taken at the right time, with as little manual effort from the user as possible.

Mobile reminder applications are not a new idea, and a handful of them, Medisafe and MyTherapy among the more recognizable names, already do a reasonable job of scheduling alerts. What is missing from most of them is a faster way to get a new medicine into the system in the first place. Typing in a name, a dosage, and a schedule every time a new prescription shows up is tedious enough that people stop bothering after the first few entries. Adding an AI-assisted scanning option, where a quick photo of the packaging fills in most of the form automatically, was one of the more direct ways to reduce that friction and increase the odds that someone keeps using the app long enough for it to matter.

1.2 Problem Statement

A large share of patients struggle to stay consistent with prescribed medication schedules, not because they lack the discipline to follow through, but because they have no dependable system reminding them when a dose is due. This leads to doses being missed outright, taken later than intended, or taken in the wrong order relative to meals or other medicines, all of which weaken the treatment outcomes the medicine was prescribed to achieve in the first place. What is needed is a mobile application that is simple enough to set up in under a minute, reliable enough that notifications actually arrive on time, and flexible enough to support both manual entry and a faster, AI-assisted alternative for people who would rather scan a label than type one out.

1.3 Goals and Objectives

The central goal of this project is to help users stay consistent with their medication schedules through reminders that are both automated and dependable. Beyond that core function, the project sets out to achieve the following objectives:

- Provide secure registration and login so that each user's reminder data stays private to their own account.

- Let users schedule medicine reminders by specifying dosage, frequency, and duration with a minimal number of input steps.
- Deliver reminders through both push notifications and voice alerts, so the method of being reminded fits how the user actually wants to be reached.
- Reduce the effort needed to add a new medicine by offering an AI-assisted scan that reads the medicine name directly off the packaging.
- Maintain a history of past reminders and adherence so users (and, indirectly, DVAGO) can see how consistently doses are being taken over time.

Taken together, these objectives are meant to make the day-to-day experience of managing medicine noticeably less effortful than it would be without the app, while keeping the underlying system simple enough to build, test, and maintain within the timeframe of a final year project.

1.4 Project Scope

The scope below sets out what the Smart Medicine Reminder System will and will not deliver in its current form, so that the team stays focused during development and anyone reviewing the project knows exactly what to expect from it.

Category	In Scope (Will be delivered)	Out of Scope (Will not be delivered)
Core Modules	User authentication, medicine reminder scheduling, push and voice notifications, AI-assisted medicine scanning, reminder/adherence history	Online medicine ordering, payment processing, prescription validation
User Roles	End User (Patient)	Admin / DVAGO staff dashboard (planned for a future phase)
Reporting & Analytics	On-screen adherence statistics: on-time rate, streaks, per-medicine breakdown	Predictive health analytics or clinical risk scoring
Scalability & Performance	Supports everyday individual usage with a responsive, low-latency experience	Enterprise-scale concurrent load testing or load-balanced production clusters
Platforms & Tech	Mobile application (Android) built with React Native	Native platform-specific rewrites; desktop or web client
Security & Compliance	Firebase Authentication, hashed credentials, HTTPS/TLS in transit	Formal certifications such as HIPAA or ISO 27001
Timeline & Deliverables	Working mobile application, SRS, SDS, test plan and report, user-facing demo	Direct integration with DVAGO's live production systems

Assumptions

- Users own an Android smartphone capable of running the application.
- Users are willing to grant the permissions the app needs, namely notifications, camera access, and audio output.
- A working internet connection is available whenever the app needs to reach Firebase or other cloud services.

-
- Users are engaging with the app primarily to manage their own medicine reminders rather than for unrelated purposes.

Dependencies

- Firebase services need to remain available and reliable, since both authentication and notification delivery run through them.
- The OCR service used for AI-assisted scanning needs to perform reasonably well at reading text off medicine packaging under normal lighting conditions.
- Push notification delivery depends on the underlying mobile operating system cooperating with Firebase Cloud Messaging.

CHAPTER 2

RELEVANT BACKGROUND & DEFINITIONS

Medication non-adherence is a well-documented problem in healthcare research, and it is worth grounding this project in that broader context before getting into the specifics of the system itself. The World Health Organization has reported that adherence to long-term therapies in developed countries averages around 50 percent, with the figure typically lower still in developing regions, where access to consistent reminders and follow-up care is more limited. This is not a niche issue confined to elderly patients or complex treatment regimens; it shows up across nearly every demographic and condition where a medicine has to be taken on a recurring schedule rather than once.

Mobile health applications have become one of the more practical responses to this problem, largely because smartphones are already in most people's pockets and do not require any additional hardware to function as a reminder tool. Two existing applications are particularly relevant to the direction taken in this project.

Medisafe

Medisafe is a medication management application that lets users log their prescriptions, set reminder schedules, and track whether doses were taken on time. It also supports caregiver notifications, where a family member can be alerted if a dose is missed. Its core strength lies in a clean, accessible interface that walks users through setting up a new medicine with minimal friction.

MyTherapy

MyTherapy follows a similar model, combining reminder scheduling with adherence tracking and a built-in health diary where users can log symptoms or vitals alongside their medicine intake. It places a fair amount of emphasis on visualizing adherence trends over time, which gives users (and, where relevant, their doctors) a clearer picture of how consistently a treatment plan is actually being followed.

Both of these applications rely almost entirely on manual data entry when a new medicine is added. There is no meaningful use of computer vision or text recognition to speed up that step, which means every new prescription requires the user to type in the name, dosage, and schedule by hand. The Smart Medicine Reminder System for DVAGO builds on the same underlying premise, automated, scheduled reminders that reduce the cognitive load of remembering a medication routine, but takes a different approach to onboarding a new medicine by introducing AI-assisted optical character recognition. A user can photograph the packaging directly, and the system extracts the medicine name from that image rather than asking the user to type it. This is a relatively small addition in terms of scope, but it is meant to noticeably lower the barrier to actually setting up and maintaining a reminder, particularly for users managing several medicines at once.

CHAPTER 3

LITERATURE REVIEW & RELATED WORK

Literature Review

- [1] JMIR, “Improving Medication Adherence with Mobile Apps,” Journal of Medical Internet Research, 2022.
- [2] A. Morrison et al., “Mobile Health Applications and Patient Self-Management,” Journal of Medical Internet Research, 2021.
- [3] World Health Organization, Adherence to Long-term Therapies: Evidence for Action, 2019.
- [4] World Health Organization, Report on Digital Health in Developing Countries.
- [5] Medisafe, “Medisafe – Medication Reminder & Pill Tracker,” [Online]. Available: <https://www.medisafe.com>. [Accessed 15 May 2026].
- [6] MyTherapy, “MyTherapy – Medication Reminder & Health Tracker,” [Online]. Available: <https://www.mytherapyapp.com>. [Accessed 15 May 2026].

Related Work

Research into medication adherence consistently points to forgetfulness and the absence of a structured reminder system as the leading reasons patients deviate from prescribed schedules, more so than cost or side effects in most studied populations. This finding shaped an early design decision for this project: the system needed to prioritize getting a reminder set up quickly over offering an extensive set of clinical features that a patient would rarely use.

Medisafe and MyTherapy remain the two most relevant points of comparison, since both address the same underlying problem with a similar reminder-and-tracking model. Medisafe's caregiver-notification feature is a thoughtful addition for patients who may need a second layer of accountability, and MyTherapy's adherence visualizations do a good job of making long-term trends easy to read at a glance. Where both fall short, relative to the goals of this project, is in how a new medicine actually gets entered into the system. Manual data entry is the only option in either app, which works fine for a single recurring prescription but becomes a noticeably bigger task for someone managing several medicines with different schedules.

The Smart Medicine Reminder System for DVAGO takes the same core reminder-and-notification model used by these platforms and extends it with an AI-assisted scanning option built around optical character recognition. Rather than typing in a medicine name, dosage, and schedule from scratch, a user can photograph the medicine packaging and have the system extract the relevant text automatically, with the option to manually correct or complete any details the scan does not pick up cleanly. This does not replace manual entry, which remains available as a fallback, but it does meaningfully shorten the time it takes to add a new medicine, particularly for users who are less comfortable typing on a small mobile keyboard.

Gap Analysis

Neither Medisafe nor MyTherapy offers an AI-assisted way to populate medicine details from a scanned image. Both rely entirely on the user typing in the relevant fields, which is a reasonable approach for a single medicine but becomes repetitive and time-consuming as the number of active prescriptions grows. This gap is where the Smart Medicine Reminder System for DVAGO differentiates itself, by combining the same dependable, scheduled reminder model used by existing apps with a faster onboarding path powered by optical character recognition.

Feature	Medisafe	MyTherapy	Smart Medicine Reminder (DVAGO)
AI-Assisted Medicine Entry	Not available	Not available	Integrated via OCR-based scanning of medicine packaging
Push Notifications	Available	Available	Available, with configurable sound and vibration
Voice Alerts	Not available	Not available	Available as a configurable reminder option
Adherence Tracking & Analytics	Available	Available, with detailed visual trends	Available, with weekly/monthly/yearly views and per-medicine breakdown
Target Platform	Android and iOS	Android and iOS	Android (built with React Native)

Table 3: Gap Analysis

CHAPTER 4

PROJECT DISCUSSION

1. Software Engineering Methodology

The Smart Medicine Reminder System for DVAGO was developed using the Agile Software Development methodology. Agile was chosen specifically because the requirements for a reminder and notification system tend to shift as soon as real testing begins, timing tolerances, notification wording, and even the exact fields needed for a reminder all evolved as the team worked through early prototypes. Short, regular development cycles made it possible to adjust the AI scanning flow, refine the notification logic, and add features such as voice alerts and adherence analytics without having to restart planning from scratch each time a change was needed.

2. Project Methodology

At the project level, development followed an Incremental Model, where the system was built and delivered in functional pieces rather than all at once. Authentication was built and tested first, followed by reminder scheduling, then notification delivery, and finally the AI-assisted scanning module and analytics dashboard. Building the system this way meant that a working, testable version of the app existed at nearly every stage of development, which made it considerably easier to catch integration issues early rather than discovering them all at once during final testing.

3. Phases of Project

The project was carried out across the following phases:

Requirement Gathering

The team reviewed existing medication reminder applications, primarily Medisafe and MyTherapy, to understand how similar problems had already been solved and where the more obvious gaps remained. This research, combined with discussions with the project supervisors, shaped the final list of functional and non-functional requirements documented in the Software Requirements Specification, covering authentication, reminder scheduling, AI-based medicine recognition, and notification delivery.

System Design

The system architecture, database schema, and the flow of data between the mobile front end and backend services were designed and documented in the Software Design Specification. The database was modeled around three core entities, User, Reminder, and Scan, implemented on a cloud-hosted MongoDB instance, with Firebase handling authentication and push notification delivery.

Development

The mobile application was built using React Native so that a single codebase could target Android. Backend logic, including reminder validation and scheduling, was implemented in Node.js, with Firebase Cloud Functions handling background automation. The AI-assisted scanning feature was integrated using the Space OCR API to extract medicine names from photographed packaging.

Testing & Debugging

Each module, login and sign-up, reminder creation, the AI scanner, and notification delivery, was tested individually before being integrated and tested as part of the full application. Issues identified during this phase included inconsistent OCR results under poor lighting, minor delays in notification delivery on certain devices, and a handful of form-validation edge cases, all of which were resolved before the final test cycle documented in Chapter 6.

Deployment

The application was deployed for testing using Firebase for authentication, real-time database operations relevant to notification scheduling, and Firebase Cloud Messaging for push delivery. MongoDB Atlas was used as the primary data store for user profiles, reminders, and scan records.

4. Software / Tools Used in Project

Frontend

- React Native (JavaScript)

Backend

- Node.js
- Firebase Cloud Functions

Database and Cloud Services

- MongoDB Atlas
- Firebase Authentication
- Firebase Cloud Messaging (FCM)

AI Tools & Libraries

- Space OCR API (for AI-assisted medicine recognition)
- Web Speech API (for voice alert playback)

Others

- Visual Studio Code (Code Editor)
- Postman (API Testing)
- Git & GitHub (Version Control)

5. Hardware Used in Project

Development Machine

- Processor: Intel Core i5, 10th Generation
- RAM: 8 GB
- Storage: SSD 256 GB

- Operating System: Windows 10

Test Devices

- Android smartphone (for live notification, camera, and voice alert testing)

CHAPTER 5

IMPLEMENTATION

1. Proposed System Architecture / Design

The Smart Medicine Reminder System follows a layered, client-server architecture that separates the mobile front end, backend services, cloud infrastructure, and the AI-based OCR processing into distinct, loosely coupled layers. This separation keeps each part of the system easier to test, debug, and extend independently of the others.

System Level Architecture

- Mobile Application Layer: the interface users interact with to manage reminders and run scans.
- Backend Services Layer: handles authentication, reminder scheduling, and the logic behind notifications.
- Cloud Services Layer: Firebase Authentication, Firebase Cloud Messaging, and MongoDB Atlas.
- OCR Processing Layer: the Space OCR API, used to read medicine names off photographed packaging.

Software Architecture

- User Interface Layer: the mobile UI built in React Native that users see and interact with directly.
- Application Logic Layer: schedules reminders, validates input, and triggers notifications.
- Data Access Layer: stores and retrieves user data and reminder history from MongoDB.
- Integration Layer: communicates with Firebase and the external OCR API.

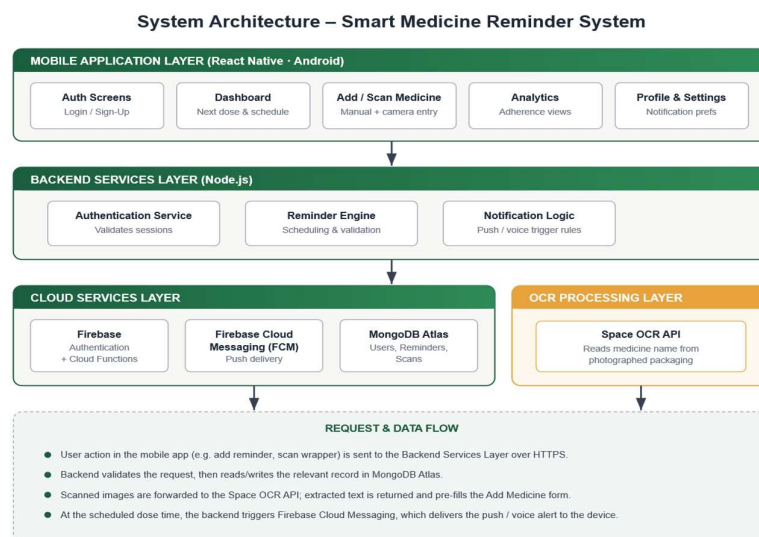


Figure 5.1: System Architecture Diagram

Figure 5.1: System Architecture Diagram

2. Functional Specifications

The functional requirements define the core operations the system is expected to perform:

- User registration and login through Firebase Authentication.
- Manual entry of medicine reminders, including dosage, frequency, duration, and reminder time(s).
- AI-assisted medicine recognition through camera-based scanning of medicine packaging.
- Scheduling and delivery of push notifications and voice alerts at the configured reminder time.
- A dashboard showing the next upcoming dose with a live countdown, and the ability to mark a dose as taken.
- A searchable, filterable list of all active medicines (My Medicines).
- An in-app notification and activity log recording dosage reminders and doses marked as taken.
- An analytics dashboard summarizing adherence rate, on-time rate, streaks, and per-medicine breakdown across week, month, and year views.
- Profile management, including editable personal details and configurable notification settings.

Use-Case Diagram

The use-case diagram below captures the primary interactions between the End User and the system. The End User is the sole active actor in the current scope of the project, responsible for registering and logging in, adding medicines either manually or through the AI scanner, viewing and managing the medicine list, receiving and acting on reminders, reviewing adherence analytics, and managing their own profile and notification preferences.

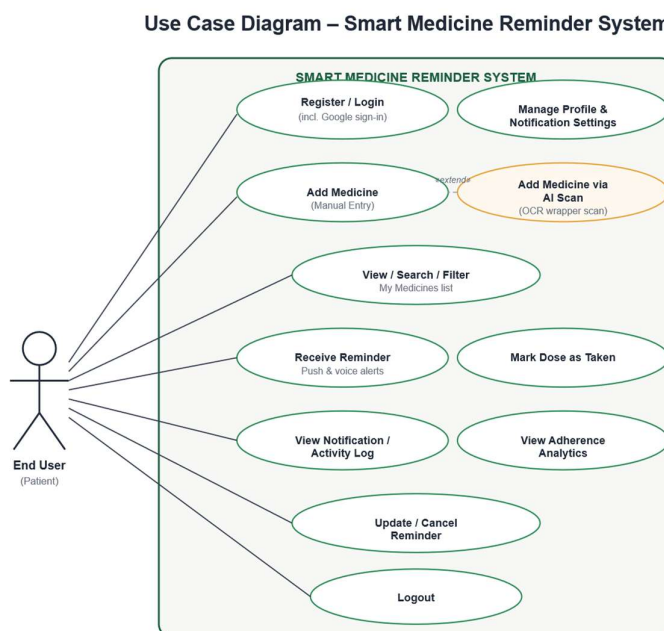


Figure 5.2: Use Case Diagram

Figure 5.2: Use Case Diagram

3. Non-Functional Specifications

The following specifications describe the quality attributes the system is expected to meet:

Performance

Reminder creation, updates, and retrieval are expected to complete within 2 seconds under normal usage conditions, and scheduled notifications should fire within a tolerance of roughly ± 5 seconds of their configured time.

Scalability

The use of Firebase and a cloud-hosted MongoDB instance allows the system to scale its user base without requiring structural changes to the application, supporting a minimum of 100 concurrent users without a noticeable drop in performance.

Security

- Firebase Authentication for secure registration and login.
- HTTPS/TLS encryption for all data transmitted between the mobile app and backend services.
- Input validation and session management to prevent malformed or unauthorized requests.

Usability

The interface is designed to remain simple and readable, with a minimal number of steps required to add a new reminder, so that the app stays approachable for users with varying levels of comfort with mobile technology.

Reliability

Reminder schedules and user data are designed to persist correctly across app restarts and device reboots, and the system is built to recover automatically from minor failures without losing previously stored reminders.

Maintainability

A modular code structure, with reminders, notifications, and AI scanning implemented as independent components, keeps the system easier to update, debug, and extend with future features such as an admin dashboard or wearable device integration.

Availability

The system is intended to eventually run on a live server with greater than 99 percent uptime, supported by Firebase's underlying cloud infrastructure.

4. Detailed System Design

4.1 Design Class Diagram

The class diagram below captures the core entities of the system and the relationships between them. It reflects the system's structure as currently implemented, and is expected to be refined further as the application continues to evolve.

User

Represents the end user (the patient) who uses the app to manage medicine reminders and notification preferences.

- Attributes: user_id, name, email, password, fcmToken
- Methods: register(), login(), updateProfile()
- Interactions: a User can create multiple Reminders and perform multiple Scans.

Scan

Represents an AI-assisted scan of a medicine wrapper, used to extract medicine information from a photographed image.

- Attributes: scan_id, medicineName, extractedText, imageUrl, createdAt
- Methods: scanImage(), saveScan()
- Interactions: each Scan belongs to one User.

Reminder

Handles the scheduling and tracking of medicine intake reminders, including whether push and voice alerts are enabled for that reminder.

- Attributes: reminder_id, medicineName, dosage, time, frequency, duration, notification, voiceAlert, notified
- Methods: scheduleReminder(), updateReminder(), cancelReminder()
- Interactions: each Reminder belongs to one User and triggers notifications through the NotificationManager.

NotificationManager

A service class responsible for managing both push and voice notifications based on each reminder's schedule.

- Attributes: notification_id, type, delivery_status
- Methods: sendPushNotification(), playVoiceAlert(), scheduleNotification()
- Interactions: receives trigger events from a Reminder and delivers the resulting alert to the User.

Class Diagram – Smart Medicine Reminder System

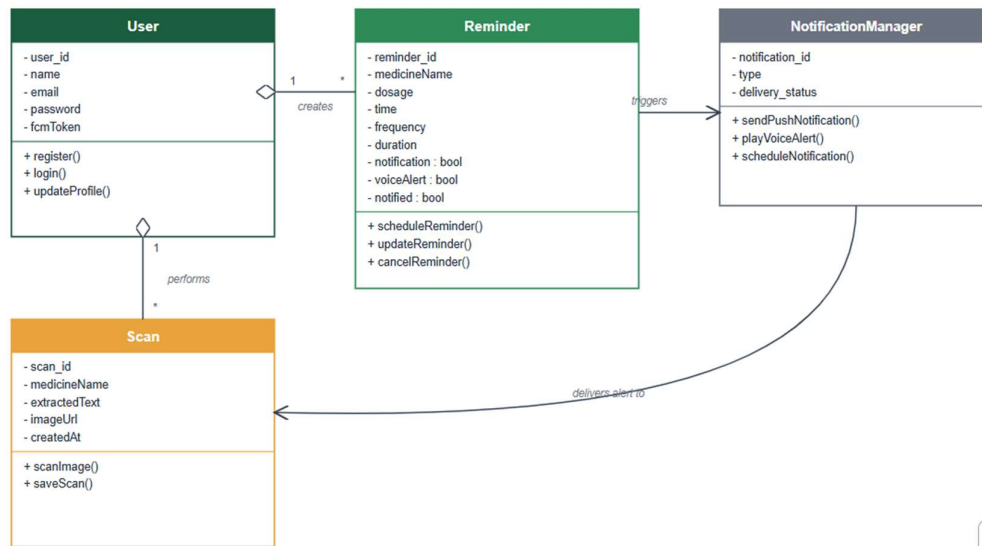


Figure 5.3: Class Diagram

4.2 Database Design

The database follows a logical Entity Relationship (ER) model implemented on a cloud-hosted MongoDB instance. This provides flexible document storage and real-time synchronization, both useful for a mobile app that needs reminder and scan data to stay consistent across sessions and devices.

- Entities: User, Reminder, Scan
- A User can create multiple Reminders.
- A User can perform multiple Scans.
- Each Reminder belongs to one User.
- Each Scan belongs to one User.

ER Diagram – Smart Medicine Reminder System

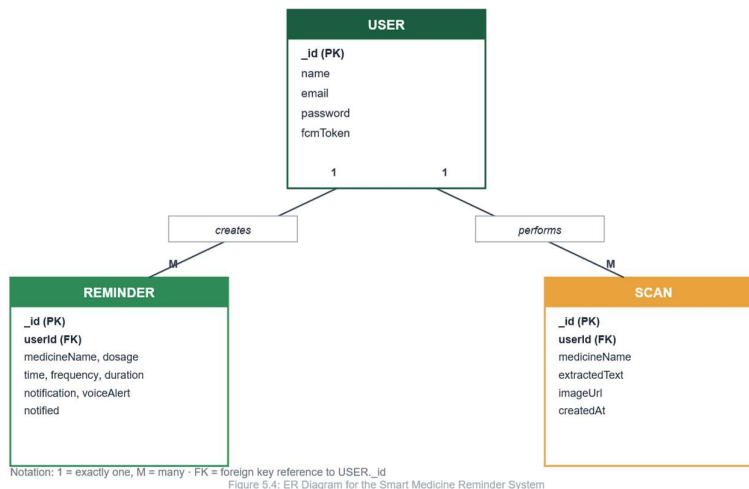


Figure 5.4: ER Diagram for the Smart Medicine Reminder System

Figure 5.4: ER Diagram for the Smart Medicine Reminder System

Table 4 below summarizes the User collection schema.

Column	Type	Length	Nullable	Default	Key
_id	ObjectId	12 bytes	No	Auto-generated	PK
name	String	255	No	NULL	
email	String	255	No	NULL	
password	String (hashed)	255	No	NULL	
fcmToken	String	255	Yes	Auto-generated	

Table 4: Data Dictionary – User

Column	Type	Length	Nullable	Default	Key
_id	ObjectId	12 bytes	No	Auto-generated	PK
userId	ObjectId	12 bytes	No	NULL	FK
medicineName	String	255	No	NULL	
dosage	String	100	No	NULL	
time	DateTime	N/A	No	NULL	
frequency	Number	N/A	No	1	
duration	Number	N/A	No	NULL	
notification	Boolean	N/A	Yes	true	
voiceAlert	Boolean	N/A	Yes	false	
notified	Boolean	N/A	Yes	false	

Table 5: Data Dictionary – Reminder

Column	Type	Length	Nullable	Default	Key
_id	ObjectId	12 bytes	No	Auto-generated	PK
userId	ObjectId	12 bytes	No	NULL	FK
medicineName	String	255	Yes	NULL	
extractedText	String	N/A	Yes	NULL	
imageUrl	String	500	No	NULL	
createdAt	Timestamp	N/A	No	CURRENT_TIMESTAMP	

Table 6: Data Dictionary – Scan

4.3 Application Design – Sequence Diagrams

The sequence diagrams below describe the flow of interaction between the End User, the mobile application, the backend server, the MongoDB database, and Firebase Cloud Messaging when a new reminder is created, covering both the manual entry path and the AI-scan path.

- End User: launches the app, chooses Manual Entry or AI Scan, enters or confirms medicine details, and receives confirmation once the reminder is created.
- Mobile Application: displays the appropriate input form or activates the camera, sends the entered or extracted data to the backend, and displays a confirmation or error message in response.
- Backend Server: receives the reminder data, validates the medicine information and scheduling details, stores the reminder, and triggers notification scheduling through Firebase Cloud Messaging.
- MongoDB Atlas: stores the validated reminder record and keeps it available for retrieval when the notification is due.
- Firebase Cloud Messaging: schedules the push or voice notification based on the dosage time and delivers the alert when that time arrives.

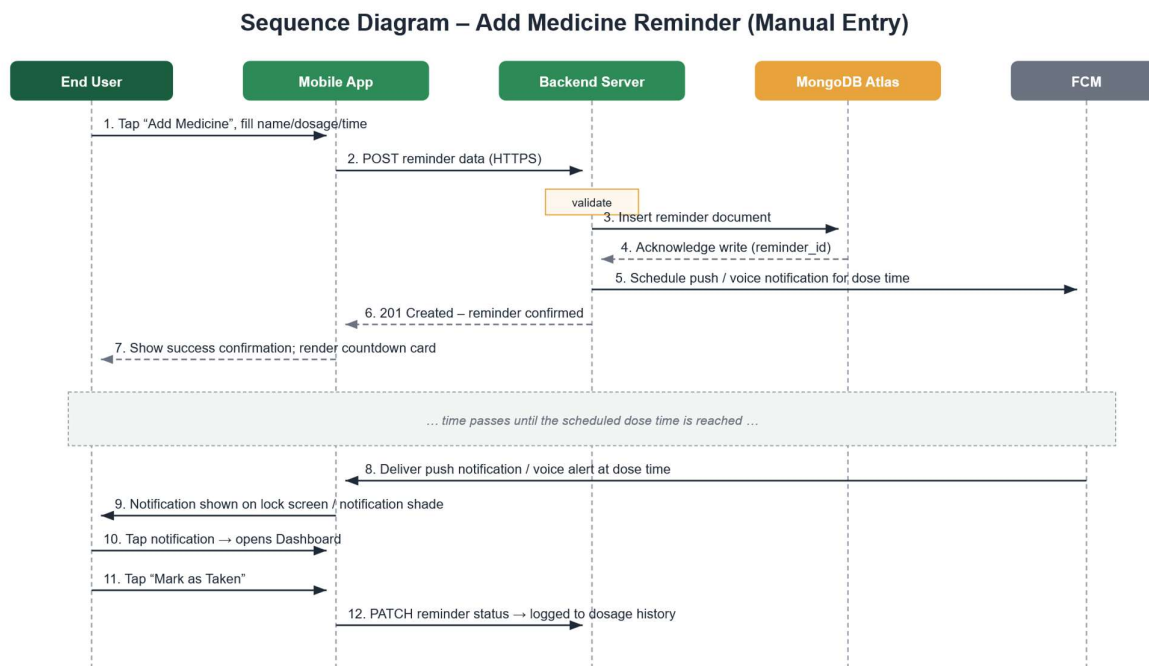


Figure 5.5: Sequence Diagram – Add Medicine Reminder (Manual Entry)

Figure 5.5: Sequence Diagram – Add Medicine Reminder (Manual Entry)

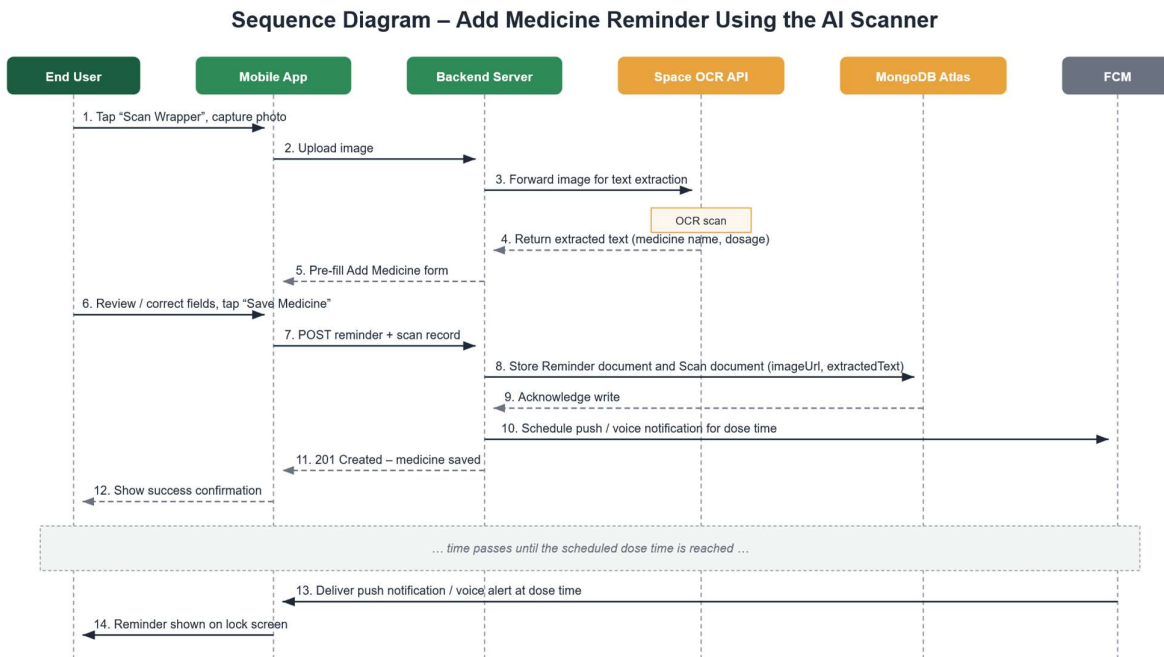


Figure 5.6: Sequence Diagram – Add Medicine Reminder Using the AI Scanner

Figure 5.6: Sequence Diagram – Add Medicine Reminder Using the AI Scanner

4.4 State Diagram

The state diagram below lays out the functional architecture of the application and how its modules, Reminder Management, User Management, AI Scanning & Processing, Notification Scheduling, and Backend Processing, work together, all backed by a shared data storage and retrieval layer that persists user profiles, medicine details, and reminder history.

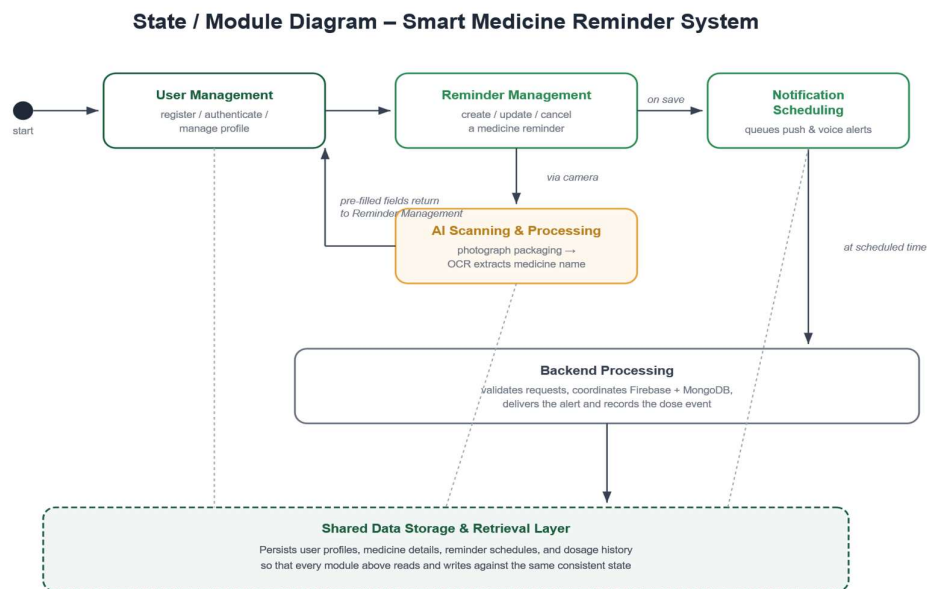


Figure 5.7: State Diagram for the Smart Medicine Reminder System

Figure 5.7: State Diagram for the Smart Medicine Reminder System

4.5 GUI Design

The screens below represent the application's working interface as implemented and tested. The general flow moves from a splash and onboarding sequence, through login and sign-up, into a dashboard that surfaces the next due dose, with dedicated screens for adding or scanning a medicine, reviewing the medicine list, checking adherence analytics, and managing the user's profile and notification settings.

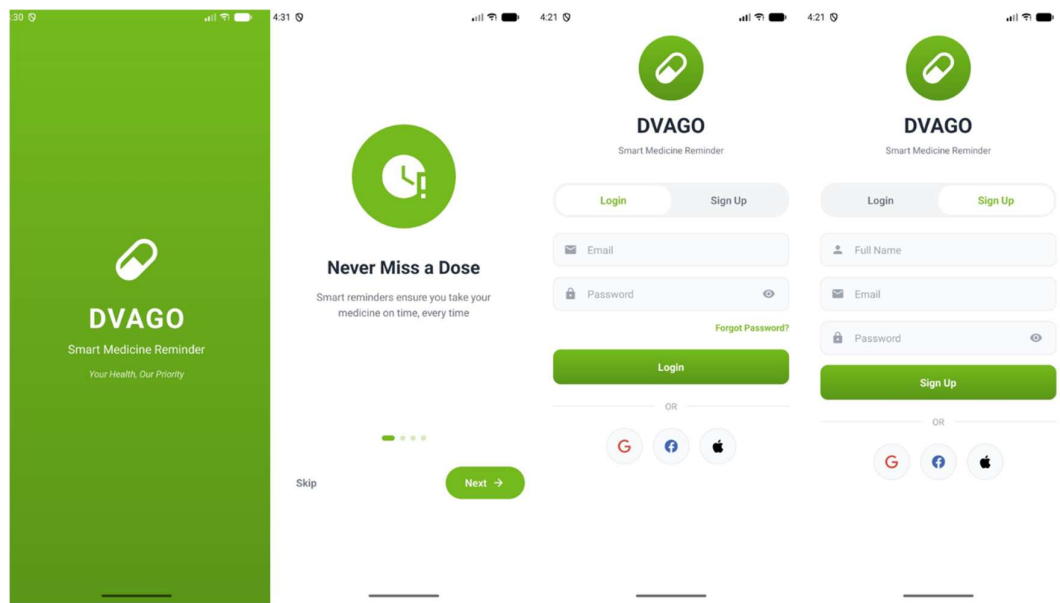


Figure 5.8: Splash, Onboarding, Login, and Sign-Up Screens

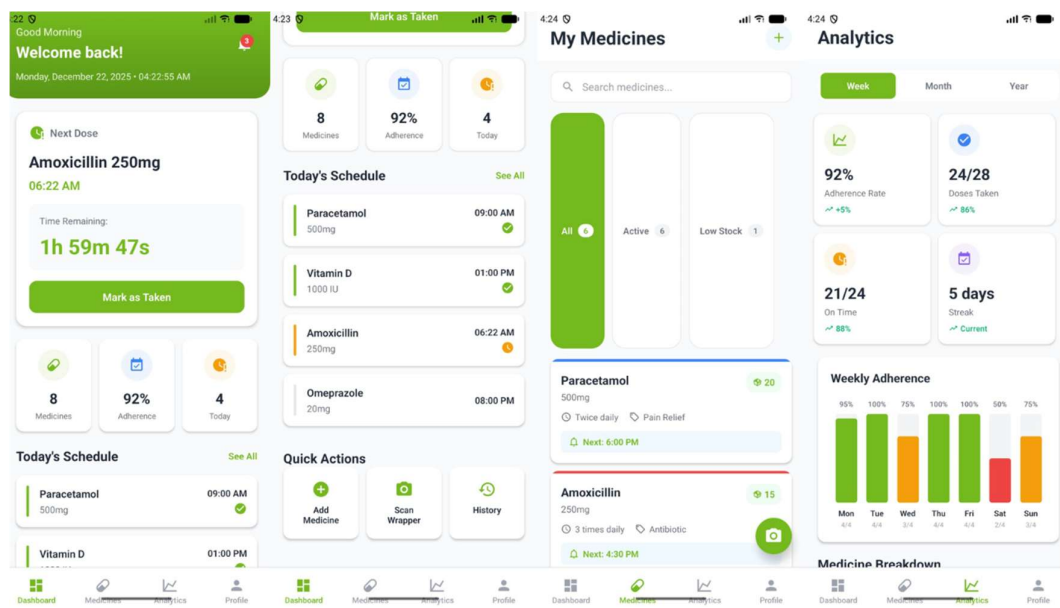


Figure 5.9: Dashboard, My Medicines, and Analytics Screens

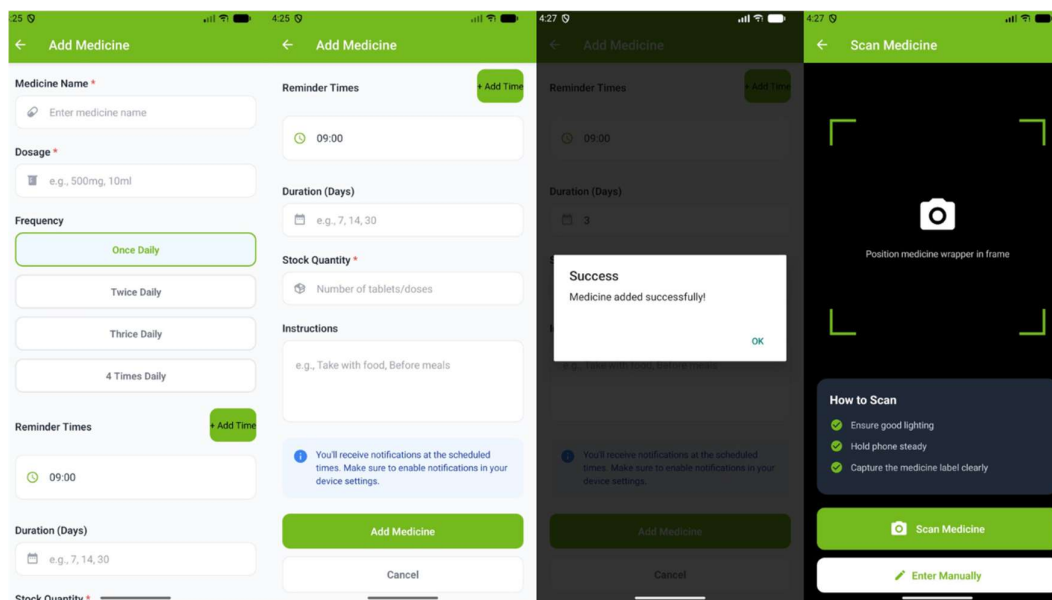


Figure 5.10: Add Medicine and Scan Medicine Screens

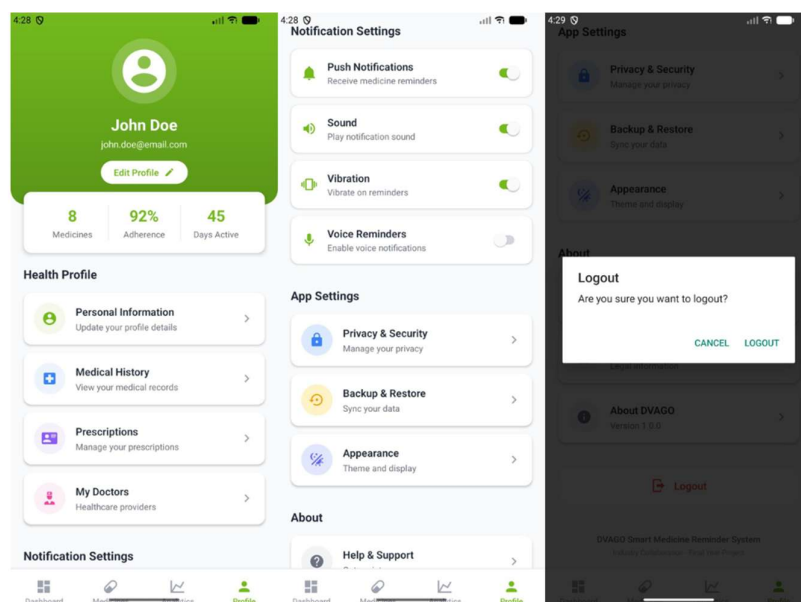


Figure 5.11: Profile, Notification Settings, and App Settings Screens

5. Testing

Testing was carried out in multiple phases to confirm the system's reliability and overall functionality.

Unit Testing

Each module, including login, reminder scheduling, and the AI scanner, was tested individually before being integrated with the rest of the system.

Integration Testing

Verified the flow of data between the mobile front end, backend logic, Firebase, and MongoDB, confirming that reminders created on the app were correctly stored and retrievable.

System Testing

End-to-end testing of complete workflows, including registration, adding a medicine through both manual entry and the AI scanner, receiving a notification, and reviewing adherence analytics.

User Acceptance Testing (UAT)

Conducted with real users to confirm the application felt usable, intuitive, and dependable in practice, not just in isolated test conditions.

6. Purpose of Testing

The purpose of the testing effort was to:

- Identify and resolve bugs or logic errors before final delivery.
- Confirm that the system meets the functional and non-functional requirements set out in the SRS.
- Validate that core user journeys, adding a medicine, receiving a reminder, marking a dose as taken, are seamless and behave as expected.
- Check the accuracy and reliability of the AI-based medicine scanning feature.
- Ensure notification delivery remains consistent across different devices and lock-screen states.

CHAPTER 6

EXPERIMENTAL EVALUATIONS & RESULTS

1. Evaluation Testbed

To evaluate the system's effectiveness and overall reliability, a controlled test environment was set up covering the application's core functional areas: authentication, reminder scheduling, the AI-assisted scanning module, notification delivery, and adherence analytics.

Development Machine Specifications

- Processor: Intel Core i5, 10th Generation
- RAM: 8 GB
- Operating System: Windows 10
- Browsers / Tools: Chrome (for backend and API verification)

Testing Tools

- Postman (API testing)
- Firebase Console (backend and notification monitoring)
- Visual Studio Code (development and debugging)
- Physical Android device and Android Emulator (UI, camera, and notification testing)

Test Coverage

Ten test cases were executed across the application's major modules, covering login and sign-up with network-retry behaviour, onboarding and notification permissions, the dashboard's next-dose countdown, manual medicine entry, the AI-based wrapper scan, the My Medicines list with search and filters, lock-screen dose reminders, the in-app notification log, the analytics dashboard, and profile management with logout. The full test plan and detailed test cases are documented in Appendix A4.

2. Results and Discussion

2.1 Functional Test Results

All ten test cases passed across every step executed. Login and sign-up completed correctly, including a successful recovery from a simulated network error, and the onboarding flow correctly requested notification permissions before walking the user through a short introductory carousel. The dashboard reliably displayed the next upcoming dose with a live, ticking countdown and allowed doses to be marked as taken, after which the next scheduled medicine populated correctly.

Manual medicine entry accepted all required fields, dosage, frequency, reminder time, duration, stock quantity, and instructions, and produced a working countdown card immediately after submission. The AI-based wrapper scan correctly detected the medicine name and dosage from a photographed package in the tested scenario, pre-filling the Add Medicine form while still allowing the remaining fields to be completed manually.

Test Case ID	Module	Steps Passed	Result
TC_001	Login and Sign-Up	4 / 4	Pass
TC_002	Onboarding & Notification Permission	4 / 4	Pass
TC_003	Dashboard – Next Dose & Mark as Taken	5 / 5	Pass
TC_004	Add Medicine (Manual Entry)	7 / 7	Pass
TC_005	Scan Medicine (AI Wrapper Scan)	6 / 6	Pass
TC_006	My Medicines – List, Search & Filters	6 / 6	Pass
TC_007	Dose Reminder Notification (Lock Screen)	4 / 4	Pass
TC_008	Notifications Panel & Activity Log	4 / 4	Pass
TC_009	Analytics Dashboard (Adherence Tracking)	6 / 6	Pass
TC_010	Profile Management & Logout	6 / 6	Pass

Table 7: Summary of Functional Test Results

2.2 Notification Reliability

Lock-screen testing confirmed that scheduled dose reminders fired correctly at the configured time, remained visible in the notification shade, and opened directly to the relevant dashboard view when tapped. The in-app notification log correctly recorded both dosage reminders and dose-taken events with accurate timestamps, and the unread badge count matched the number of unread entries throughout testing, including after clearing the log.

2.3 Adherence Analytics

The analytics dashboard correctly recalculated adherence statistics when switching between week, month, and year views. During the test cycle, the week view reported an 83 percent on-time rate with a one-day streak, while the month view reported an 86 percent adherence rate across six of seven scheduled doses. Per-medicine breakdowns and the system's narrative insights, covering adherence rate, on-time rate, current streak, and best performance time of day, were consistent with the underlying logged data.

Metric	Observed Value
On-Time Rate (Week)	83%
Adherence Rate (Month)	86%
Doses Taken (Month)	6 / 7
Current Streak	1 day
Best Performance Time	Morning

Table 8: Adherence Analytics – Sample Test Period

2.4 Discussion

The results confirm that the Smart Medicine Reminder System performs reliably across its core workflows, from authentication and reminder creation through to notification delivery and adherence

reporting. The AI-assisted scanner performed well under the tested conditions, correctly extracting medicine details from a clearly photographed package, though, consistent with the risk noted in the design phase, recognition accuracy is expected to degrade under poor lighting or with a blurry or angled photo. Manual entry remains available as a dependable fallback in those cases. Firebase's notification infrastructure proved dependable throughout testing, with reminders consistently reaching the lock screen at the expected time. Taken together, these results support the conclusion that the system meets the functional and non-functional goals defined during the design phase.

CHAPTER 7

CONCLUSION AND DISCUSSION

1. Strengths of this Project

The Smart Medicine Reminder System achieves its core objective of making medication adherence less dependent on a user's own memory. Its main strengths include:

AI-Assisted Medicine Recognition

The integration of OCR-based scanning through the Space OCR API allows users to add a new medicine by photographing its packaging rather than typing every field by hand, meaningfully reducing the effort needed to set up a reminder.

Reliable, Dual-Channel Notifications

Firebase Cloud Messaging delivers push notifications consistently, and the option to enable voice alerts alongside standard notifications gives users a way to be reminded that fits their own preferences and circumstances.

Firebase and MongoDB Integration

Combining Firebase for authentication and notification delivery with MongoDB Atlas for structured data storage gives the system a scalable, real-time foundation without adding unnecessary backend complexity.

Adherence Analytics

The analytics dashboard gives users a clear, visual sense of how consistently they are taking their medicine, broken down by week, month, and year, which adds genuine value beyond simply firing reminders.

User-Friendly Interface

A clean, mobile-first interface keeps the core actions, adding a medicine, marking a dose as taken, checking the next reminder, within easy reach, which matters for an application meant to be used by people across a wide range of comfort levels with technology.

Secure Data Handling

Firebase Authentication, encrypted data transmission, and structured session handling give the system a reasonably solid security foundation for the kind of personal health-adjacent data it stores.

2. Limitations and Future Work

While the system performs well overall, a few limitations were identified during development and testing:

Limitations

- **Scan Accuracy Under Poor Conditions:** The AI scanner's accuracy depends heavily on lighting and image clarity, and can struggle with worn, faded, or unusually formatted packaging.

- **No Caregiver or Multi-User Support:** The current version is built around a single end user and does not yet support caregiver notifications or shared family accounts.
- **Limited Admin Visibility:** There is currently no admin-facing dashboard for DVAGO staff to monitor aggregate usage or system health.
- **Single-Platform Testing Emphasis:** While the application targets both Android and iOS through React Native, the bulk of device-level testing in this cycle was carried out on Android.

Future Work

- Introduce caregiver notifications so a family member can be alerted if a dose is missed.
- Build a lightweight admin dashboard for DVAGO to monitor usage trends and system health.
- Improve OCR pre-processing (image cropping and contrast adjustment) to make scanning more reliable under imperfect lighting.
- Expand iOS-specific device testing to match the depth of coverage currently completed on Android.
- Explore deeper integration with DVAGO's existing pharmacy platform, such as auto-suggesting reminders based on a recent order.

3. Reasons for Failure

The project did not experience failure in any of its core objectives. A small number of development challenges came up along the way, none of which required dropping a planned feature:

- **Notification Timing on Certain Devices:** Some Android devices delayed notifications slightly due to aggressive battery-optimization settings; this was mitigated by relying on Firebase Cloud Messaging's retry behaviour.
- **OCR Inconsistency Under Poor Lighting:** Early scan tests under dim lighting produced incomplete text extraction; this was addressed by keeping manual entry available as a dependable fallback.
- **Firebase and MongoDB Synchronization:** Coordinating real-time updates between Firebase-driven notifications and MongoDB-stored reminder data required some additional debugging early in development, but was resolved well before final testing.

Each of these issues was identified and addressed during development, and the final system met its planned functional and non-functional requirements. The project is considered successful in both its technical implementation and its alignment with the original problem statement.

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APPENDICES

List of Appendices

- A1a. Project Proposal and Vision Document
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A1A. PROJECT PROPOSAL DOCUMENT

Below is the link to the Project Proposal Document:

Link: https://drive.google.com/drive/folders/1yGqlkwhMNaDVfafuhIrFep9Rdq-64wjx?usp=drive_link

A1B. COPY OF PROPOSAL EVALUATION COMMENTS BY JURY

The jury's evaluation comments from the Project Proposal stage are attached below as a scanned document, capturing the feedback incorporated into the subsequent revisions to the project's scope and methodology.

Project Name:	Score /40	Jury Comments
Smart Medicine Reminder System for DVAGO	26	The proposed project currently lacks novelty and does not sufficiently differentiate itself from existing solutions. Since the requirements have been provided by the industry, it is important that the project reflects strong alignment with those expectations. It has been observed that the students are developing a completely standalone application, rather than integrating it effectively with the DVAGO system. To enhance the product's value and strengthen industry collaboration, it is highly recommended to incorporate a unique and impactful feature. For example, integrating an AI-based functionality such as identifying medicine names through scanning wrappers or boxes. It could significantly improve accessibility for users who may struggle to recognize medicines. Such a feature would not only add innovation but also increase the project's practical utility and appeal to a wider audience. The current project idea is not very unique and doesn't clearly stand out from other existing solutions. Since the project is based on industry requirements, it should better match what the industry expects. Right now, the students are working on a separate application instead of connecting it with the DVAGO system. To make the project more useful and relevant, it's suggested to add a creative feature, like using AI to recognize medicine names by scanning the box or wrapper. This would make the app more helpful, especially for users who have trouble identifying medicines. Thesis is ok
FYP Committee Comments:	Add features in scope of project after discussing it with internal and external supervisor.	
FYP Convener Recommendation:	Conditionally Accepted (Re-submit updated proposal document along with the compliance report by 25 July)	

A2. REQUIREMENT SPECIFICATIONS

Below is the link to the Software Requirements Specification (SRS) document:

Link:

https://drive.google.com/drive/folders/1SylDEczgxjVRk7Z8z6OfXlAZ0oZ1qdR?usp=drive_link

A3. DESIGN SPECIFICATIONS

Below is the link to the Software Design Specification (SDS) document:

Link:

https://drive.google.com/drive/folders/1gSuIMAwLFHhMNnz8D4SfYVkJSatIo7ztm?usp=drive_link

A4. OTHER TECHNICAL DETAIL DOCUMENTS

Test Plan and Test Cases Document

The following table presents the detailed test plan, including the test schedule for each functional component of the Smart Medicine Reminder System.

S.No	Description	Tested By	Start Date	End Date
1	Login and Sign-Up	Abdul Ghani	06-Jun-2026	06-Jun-2026
2	Onboarding & Notification Permission	Abdul Ghani	07-Jun-2026	07-Jun-2026
3	Dashboard – Next Dose & Mark as Taken	Abdul Ghani	08-Jun-2026	08-Jun-2026
4	Add Medicine (Manual Entry)	Abdul Ghani	10-Jun-2026	10-Jun-2026
5	Scan Medicine (AI Wrapper Scan)	Abdul Ghani	10-Jun-2026	10-Jun-2026
6	My Medicines – List, Search & Filters	Abdul Ghani	11-Jun-2026	11-Jun-2026
7	Dose Reminder Notification (Lock Screen)	Abdul Ghani	11-Jun-2026	11-Jun-2026
8	Notifications Panel & Activity Log	Abdul Ghani	11-Jun-2026	11-Jun-2026
9	Analytics Dashboard (Adherence Tracking)	Abdul Ghani	11-Jun-2026	11-Jun-2026
10	Profile Management & Logout	Abdul Ghani	11-Jun-2026	11-Jun-2026

Table 9: Summary of Software Test Plan

Test Case 1 – Login, Sign-Up & Network Retry

Field	Detail
Project Name	Smart Medicine Reminder System for DVAGO
Module Name	Login, Sign-Up
Date	06-Jun-2026
Test Case ID	TC_001
Test Engineer	Abdul Ghani
Description	Testing user login with valid credentials and verifying the Sign-Up tab is accessible.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
1	Open app and reach Login screen	–	Splash loads; Login/Sign-Up tabs displayed	Login screen loaded with both tabs	Pass
2	Enter email & password, tap Login	Valid credentials	User authenticated, redirected to Dashboard	Login succeeded, Dashboard loaded	Pass
3	Tap “Sign Up” tab	–	Sign-Up form displayed	Sign-Up tab switched correctly	Pass
4	Check “Continue with Google” option	–	Google sign-in option visible on both tabs	Google button visible and tappable	Pass

Table 10: Test Case 1 – Login, Sign-Up**Test Case 2 – Onboarding & Notification Permission**

Field	Detail
Project Name	Smart Medicine Reminder System for DVAGO
Module Name	Onboarding & Notification Permission
Date	07-Jun-2026
Test Case ID	TC_002
Test Engineer	Abdul Ghani
Description	Verify first-launch notification permission prompt and onboarding carousel.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
1	Launch app for the first time	–	System dialog requests notification permission	Permission dialog shown	Pass
2	Tap “Allow”	–	Permission granted; onboarding proceeds	Permission granted, onboarding displayed	Pass
3	Swipe through onboarding slides	–	Slides render with Skip/Next controls	Both slides rendered with pagination dots	Pass

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
4	Tap “Next” on final slide	–	Onboarding completes, Login screen shown	Redirected to Login screen	Pass

Table 11: Test Case 2 – Onboarding & Notification Permission

Test Case 3 – Dashboard – Next Dose & Mark as Taken

Field	Detail
Project Name	Smart Medicine Reminder System for DVAGO
Module Name	Dashboard – Next Dose & Mark as Taken
Date	08-Jun-2026
Test Case ID	TC 003
Test Engineer	Abdul Ghani
Description	Confirm the dashboard shows the correct upcoming dose, live countdown, and supports marking a dose as taken.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
1	Open Dashboard after login	–	Greeting banner and summary cards load	Banner and summary cards rendered correctly	Pass
2	View “Next Dose” card	–	Card shows medicine, dosage, time, countdown	Countdown displayed and ticking down correctly	Pass
3	Tap “Mark as Taken” when due	Dose due	Dose recorded; next dose loads	Card updated with next scheduled medicine	Pass
4	Review “Today’s Schedule” list	–	All medicines due today listed with status	All scheduled medicines listed with status ticks	Pass
5	Use “Quick Actions” shortcuts	–	All shortcuts visible and tappable	All quick-action tiles functional	Pass

Table 12: Test Case 3 – Dashboard – Next Dose & Mark as Taken

Test Case 4 – Add Medicine (Manual Entry)

Field	Detail
Project Name	Smart Medicine Reminder System for DVAGO
Module Name	Add Medicine (Manual Entry)
Date	10-Jun-2026
Test Case ID	TC_004
Test Engineer	Abdul Ghani

Field	Detail
Description	Verify a new medicine can be added manually with dosage, frequency, reminder times, duration, stock, and instructions.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
1	Tap “Add Medicine”	–	Add Medicine form opens	Form opened with required fields	Pass
2	Enter medicine name & dosage	Name, dosage	Fields accept input	Values entered successfully	Pass
3	Select frequency	e.g., 4 Times Daily	Frequency selection saved	Frequency saved correctly	Pass
4	Add reminder time(s)	Time via picker	Time added to Reminder Times list	Time added correctly	Pass
5	Enter duration, stock & instructions	Duration, qty, notes	Fields accept and retain input	Values saved in form	Pass
6	Tap “Add Medicine” (submit)	–	Success confirmation; medicine saved	Success dialog shown, medicine saved	Pass
7	Verify new medicine on Dashboard	–	Medicine appears with live countdown	Countdown card rendered correctly	Pass

Table 13: Test Case 4 – Add Medicine (Manual Entry)

Test Case 5 – Scan Medicine (AI Wrapper Scan)

Field	Detail
Project Name	Smart Medicine Reminder System for DVAGO
Module Name	Scan Medicine (AI Wrapper Scan)
Date	10-Jun-2026
Test Case ID	TC_005
Test Engineer	Abdul Ghani
Description	Validate the camera-based medicine wrapper scan auto-detects medicine details and pre-fills the Add Medicine form.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
1	Tap “Scan Wrapper”	–	Scan screen opens with framing guide	Instructions and framing guide shown	Pass
2	Tap “Scan Medicine” to open camera	–	Camera view opens	Camera opened in photo mode	Pass
3	Capture medicine wrapper	Medicine box	Photo captured and processed by AI	Photo captured successfully	Pass
4	Review detection result	–	Detection confirmation with auto-filled fields	Name and dosage pre-filled correctly	Pass

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
5	Complete remaining fields	Category, frequency, etc.	Fields editable after auto-fill	All fields updated and retained	Pass
6	Tap “Save Medicine”	–	Success confirmation; medicine saved to list	Medicine saved successfully	Pass

Table 14: Test Case 5 – Scan Medicine (AI Wrapper Scan)

Test Case 6 – My Medicines – List, Search & Filters

Field	Detail
Project Name	Smart Medicine Reminder System for DVAGO
Module Name	My Medicines – List, Search & Filters
Date	11-Jun-2026
Test Case ID	TC_006
Test Engineer	Abdul Ghani
Description	Confirm the My Medicines screen lists all medicines and supports search and status filters (All / Active / Low Stock).

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
1	Navigate to “Medicines” tab	–	Full list loads with search bar and filters	List loaded with filter chips and counts	Pass
2	Tap “All” filter	–	All medicines displayed	Full list displayed	Pass
3	Tap “Active” filter	–	Only active-schedule medicines shown	Active subset displayed correctly	Pass
4	Tap “Low Stock” filter	–	Only low-stock medicines shown	Low Stock subset displayed correctly	Pass
5	Use search bar	Medicine name	List narrows to matching results	Filtered results matched query	Pass
6	Open a medicine card	Selected medicine	Card expands with dosage and schedule	Details shown correctly	Pass

Table 15: Test Case 6 – My Medicines – List, Search & Filters

Test Case 7 – Dose Reminder Notification (Lock Screen)

Field	Detail
Project Name	Smart Medicine Reminder System for DVAGO
Module Name	Dose Reminder Notification (Lock Screen)
Date	11-Jun-2026
Test Case ID	TC_007

Field	Detail
Test Engineer	Abdul Ghani
Description	Verify the app fires a system notification and lock-screen alert at the scheduled dose time.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
1	Lock device before scheduled time	Scheduled dose time	Device remains locked, no premature alert	No premature alert observed	Pass
2	Wait for scheduled time to elapse	–	Reminder notification appears on lock screen	Notification displayed on lock screen	Pass
3	Pull down notification shade	–	Reminder visible and expandable in shade	Notification visible with message	Pass
4	Unlock and open app from notification	–	App opens to Dashboard showing due dose	Dashboard opened with “Due now!” status	Pass

Table 16: Test Case 7 – Dose Reminder Notification (Lock Screen)

Test Case 8 – Notifications Panel & Activity Log

Field	Detail
Project Name	Smart Medicine Reminder System for DVAGO
Module Name	Notifications Panel & Activity Log
Date	11-Jun-2026
Test Case ID	TC_008
Test Engineer	Abdul Ghani
Description	Confirm the in-app Notifications panel logs dose-taken and reminder events, and supports Clear All.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
1	Tap bell icon on Dashboard	–	Notifications panel opens	Panel opened with empty initial state	Pass
2	Trigger a reminder and mark as taken	–	Two log entries created	Both entries logged with timestamps	Pass
3	Re-open Notifications panel	–	Badge count matches unread entries	Badge count matched correctly	Pass
4	Tap “Clear All”	–	All entries removed; badge resets	List cleared, badge reset to 0	Pass

Table 17: Test Case 8 – Notifications Panel & Activity Log

Test Case 9 – Analytics Dashboard (Adherence Tracking)

Field	Detail
Project Name	Smart Medicine Reminder System for DVAGO
Module Name	Analytics Dashboard (Adherence Tracking)
Date	11-Jun-2026
Test Case ID	TC_009
Test Engineer	Abdul Ghani
Description	Validate the Analytics tab computes adherence statistics correctly across Week / Month / Year views and surfaces insights.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
1	Navigate to “Analytics” tab	–	Screen loads with view toggle	Screen loaded, Week selected by default	Pass
2	View “Week” adherence	–	Chart and stats render for the week	83% on-time rate, 1-day streak shown	Pass
3	Switch to “Month” view	–	Stats recalculate for monthly range	86% adherence rate, 6/7 doses shown	Pass
4	Switch to “Year” view	–	Stats recalculate for yearly range	Year view data refreshed correctly	Pass
5	Review “Medicine Breakdown” section	–	Per-medicine adherence percentage listed	Per-medicine bars listed correctly	Pass
6	Review “Insights” section	–	Adherence, on-time rate, streak summarised	All insight metrics displayed correctly	Pass

Table 18: Test Case 9 – Analytics Dashboard (Adherence Tracking)

Test Case 10 – Profile Management & Logout

Field	Detail
Project Name	Smart Medicine Reminder System for DVAGO
Module Name	Profile Management & Logout
Date	11-Jun-2026
Test Case ID	TC_010
Test Engineer	Abdul Ghani
Description	Confirm profile details can be viewed/edited and that logout requires confirmation and returns the user to the Login screen.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
1	Navigate to “Profile” tab	–	Profile loads with name, email, stats	Profile loaded with summary stats	Pass

S.No	Steps	Input Data	Expected Result	Actual Result	Pass / Fail
2	Tap “Edit Profile”	–	Edit dialog opens with editable fields	Dialog opened with pre-filled fields	Pass
3	Open “Health Profile” subsections	–	Subsections accessible	All subsections listed and tappable	Pass
4	Open “Notification Settings”	–	Toggles for push, sound, vibration, voice shown	All four toggles displayed and switchable	Pass
5	Tap “Logout”	–	Confirmation dialog shown	Dialog shown with Cancel/Logout options	Pass
6	Confirm “Logout”	–	Session ends, redirected to Login	Redirected to Login screen successfully	Pass

Table 19: Test Case 10 – Profile Management & Logout

End of Test Plan & Test Cases

Below is the link for the full Test Cases Document:

Link:

https://drive.google.com/drive/folders/1eGBaqO3bg3S_yXpR0UTg3JQPixbiwpLz?usp=drive_link

UI/UX Detail Document

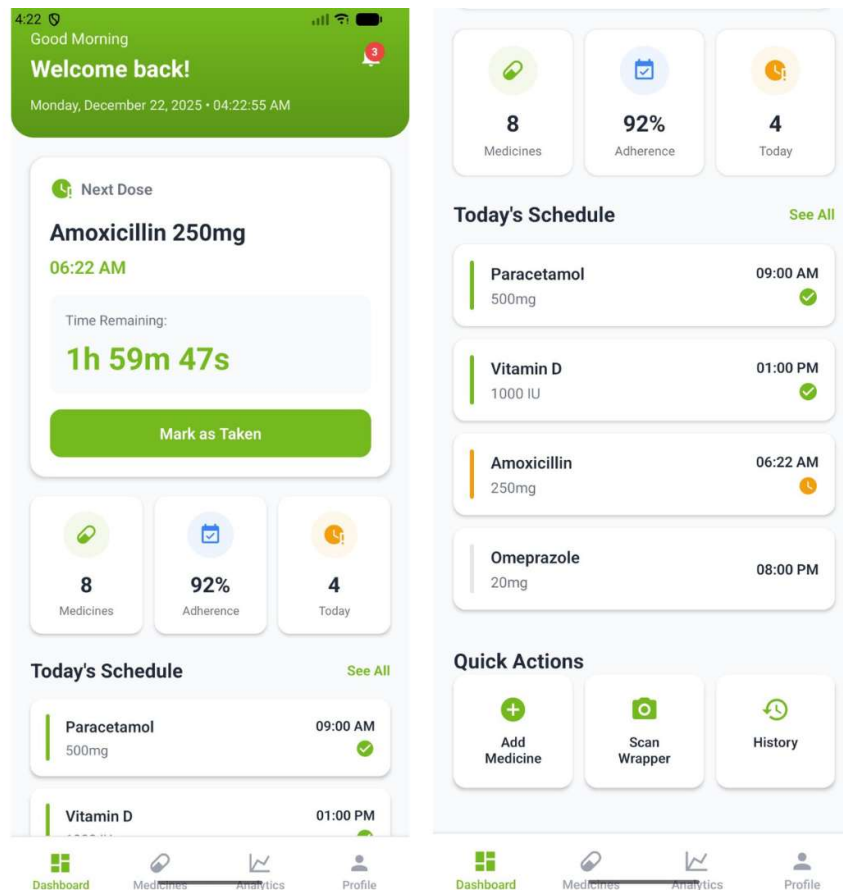
Below is the link to the UI/UX Detail Document:

Link: https://drive.google.com/drive/folders/1HaK3ZDPQgJ4WorxZflTLKqx-hGMgFqiz?usp=drive_link

A brief sample of the application's interface is provided below.

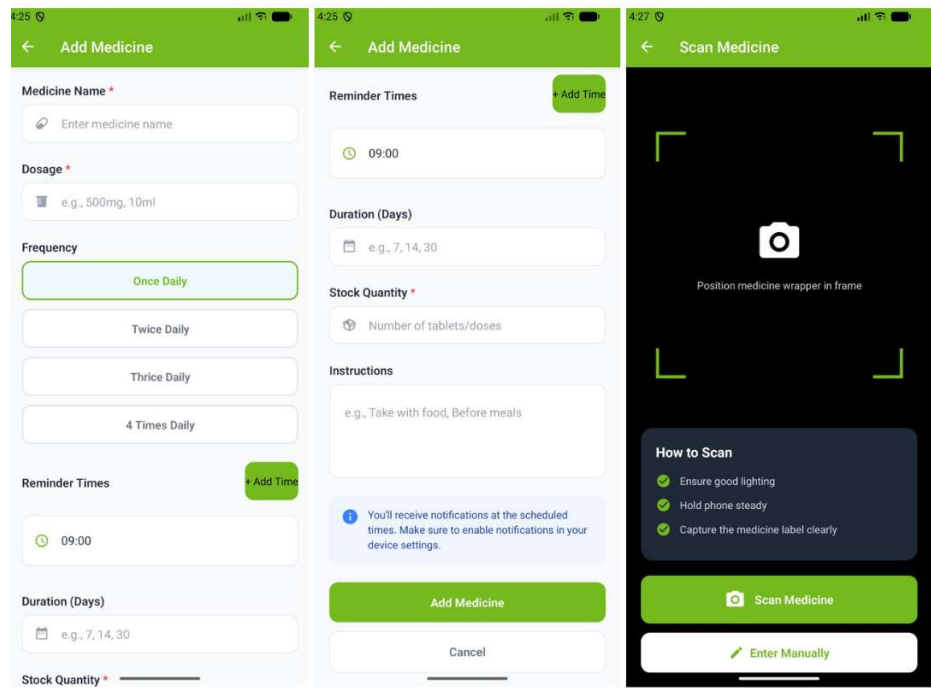
Dashboard – Next Dose & Today's Schedule

The dashboard greets the user and surfaces the medicine due next, complete with a live countdown timer, alongside a quick summary of medicines, adherence percentage, and today's remaining doses.



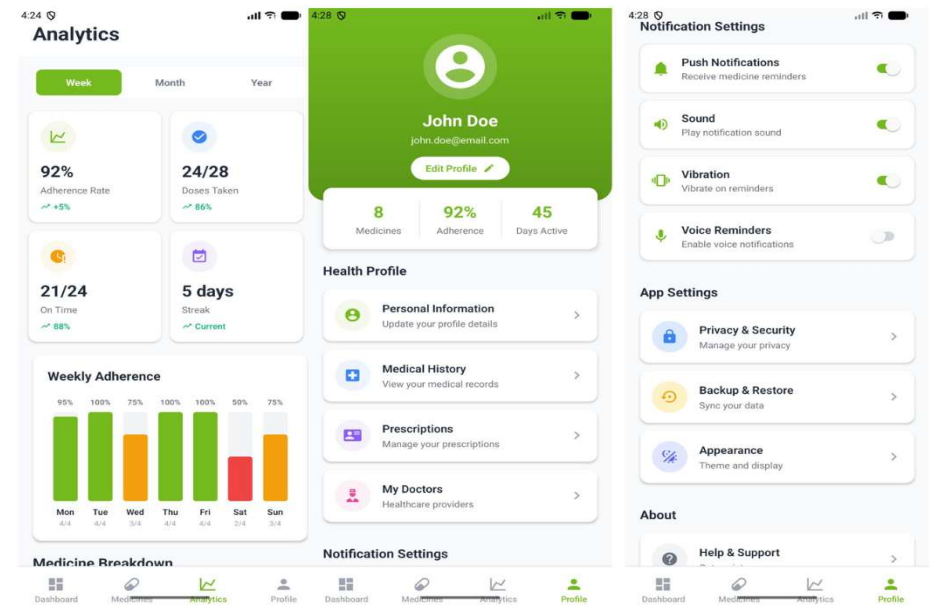
Add Medicine & Scan Medicine

Users can add a medicine manually by entering its name, dosage, frequency, and reminder times, or use the camera-based scanner, which detects the medicine name from the packaging and pre-fills the form.



Analytics & Profile

The analytics screen visualizes adherence across week, month, and year views, while the profile screen lets users edit their details and manage notification preferences.



Coding Standards Document

Below is the link to the Coding Standards Document:

Link: https://drive.google.com/drive/folders/1ZO2N8kSkdROTCnMG-3fH97KpDVyFT-C?usp=drive_link

Project Policy Document

Below is the link to the Project Policy Document:

Link: https://drive.google.com/drive/folders/1amv2o_7-fSvkL1rXzkChTNBBif4XbXkn?usp=drive_link

User Manual Document

Below is the link to the User Manual Document:

Link:
https://drive.google.com/drive/folders/1lzRWgoC6BsLWUit5Vxw3xoAflUauUP92?usp=drive_link

A5. FLYER & POSTER DESIGN

The project flyer/poster, prepared for the FYP exhibition, is shown below.

PROJECT NAME
**SMART MEDICINE
 REMINDER SYSTEM
 FOR DVAGO**

PROJECT OBJECTIVE
 BUILDING A SYSTEM SHOULD FOCUS ON
 MEDICINE TRACKING, AI-BASED BOX SCANNING,
 VOICE & PUSH NOTIFICATIONS & REMINDER
 SCHEDULING.

PROJECT FEATURES

- MEDICINE REMINDERS WITH CUSTOMIZABLE SCHEDULES
- PUSH NOTIFICATIONS & VOICE ALERTS
- AI-BASED MEDICINE SCANNER
- CLEAN & USER-FRIENDLY UI

PROJECT STATUS
 FINAL EVALUATION

SUPERVISOR
MR MUHAMMAD SALMAN

CO-SUPERVISOR
MR MAAZ AHMED

EXTERNAL SUPERVISOR
MR MUHAMMAD HAMZA

TEAM MEMBERS
ABDUL GHANI (2606-2022)
SYED ANAS ALI (2640-2022)

A6. COPY OF EVALUATION COMMENTS

COPY OF EVALUATION COMMENTS BY JURY FOR PROJECT – I END SEMESTER EVALUATION

FYP-1	FYP-037/FA25	Overall good project
FYP-1	FYP-037/FA25	Index citation is missing in report, although project and presentation as fine. Good luck

Table 20: FYP-I Evaluation, Comments of Jury

A7. MEETINGS' MINUTES & SIGN-OFF SHEET

Below is the link to all meeting minutes and the sign-off sheet for FYP-I and FYP-II:

Link:

https://drive.google.com/drive/folders/1xeSnYhrsXOk97mhMgs3KkbTFGHYmV0On?usp=drive_link

A8. DOCUMENT CHANGE RECORD

Date	Version	Author	Change Details
18/12/2025	1.0	Abdul Ghani	Initial draft – SRS and SDS
08/06/2026	2.0	Abdul Ghani	Final draft of SDS
10/06/2026	2.0	Abdul Ghani	Final draft of SRS
11/06/2026	1.0	Abdul Ghani	Software Test Plan and Test Report
13/06/2026	2.0	Abdul Ghani	Final Year Project Report

Table 21: Document Change Record

A9. PROJECT PROGRESS

FYP-I:

Below is the link to the FYP-I fortnightly sign-up sheet and progress record:

FYP Fortnightly Sign-off Sheet

Course: ☒ FYP-1 ☐ FYP-2 Project Code: FYP-037/FA25 Project Name: Smart Medicine Reminder System for DVAGO
 Group Members Names & Reg#: Abdul Ghanani (2606-2022) Syed Anas Ali (2610-2022)
 Supervisor Name: Mr. M. Salman Co-Supervisor's Name: Mr. Maaz Akmal External Supervisor: Mr. Hamza

Meeting #	Date	Agenda (Brief Statement)	Attended By (Student's Name only)	Supervisor's Sign	Co-supervisor's Sign	FYP Officer's Sign
1	1-10-25	Requirements Gathering and Project Planning	Abdul Ghanani Syed Anas Ali			
2	15-10-25	Supervisor Feedback on Requirement File	Abdul Ghanani Syed Anas Ali			
3	29-10-25	Planning UI/UX & Theme Layout	Abdul Ghanani Syed Anas Ali			
4	12-11-25	Review Feedback of Completed UI/UX Design	Abdul Ghanani Syed Anas Ali			
5	26-11-25	AI-Based Medicine Scanning Module	Abdul Ghanani Syed Anas Ali			
6	03/12/25	Planning to setup Database	Abdul Ghanani Syed Anas			
7	10/12/25	Testing Database Operations	Abdul Ghanani Syed Anas Ali			
8	17/12/25	SRS & SDS Documentation Working	Abdul Ghanani Syed Anas Ali			
9						

FYP-II:

Below is the link to the FYP-II fortnightly sign-up sheet and progress record:

FYP Fortnightly Sign-off Sheet

Course: ☐ FYP-1 ☒ FYP-2 Project Code: FYP-037/FA25 Project Name: Smart Medicine Reminder System for DVAGO
 Group Members Names & Reg#: Abdul Ghanani (2606-2022) Syed Anas Ali (2610-2022)
 Supervisor Name: Mr. M. Salman Co-Supervisor's Name: Mr. Maaz Akmal External Supervisor: Mr. Mohd Hamza

Meeting #	Date	Agenda (Brief Statement)	Attended By (Student's Name only)	Supervisor's Sign	Co-supervisor's Sign	FYP Officer's Sign
1	10-2-26	FYP-I Review	Abdul Ghanani Syed Anas Ali			
2	17-2-26	Backend System Review	Abdul Ghanani Syed Anas Ali			
3	3/3/26	Backend Finalization	Abdul Ghanani Syed Anas			
4	9/3/26	UI Issues Fixup	Abdul Ghanani Syed Anas			
5	16/3/26	Documentation Overview	Abdul Ghanani Syed Anas			
6	14-4-26	Integration & Testing Phase	Abdul Ghanani Syed Anas			
7						
8						
9						